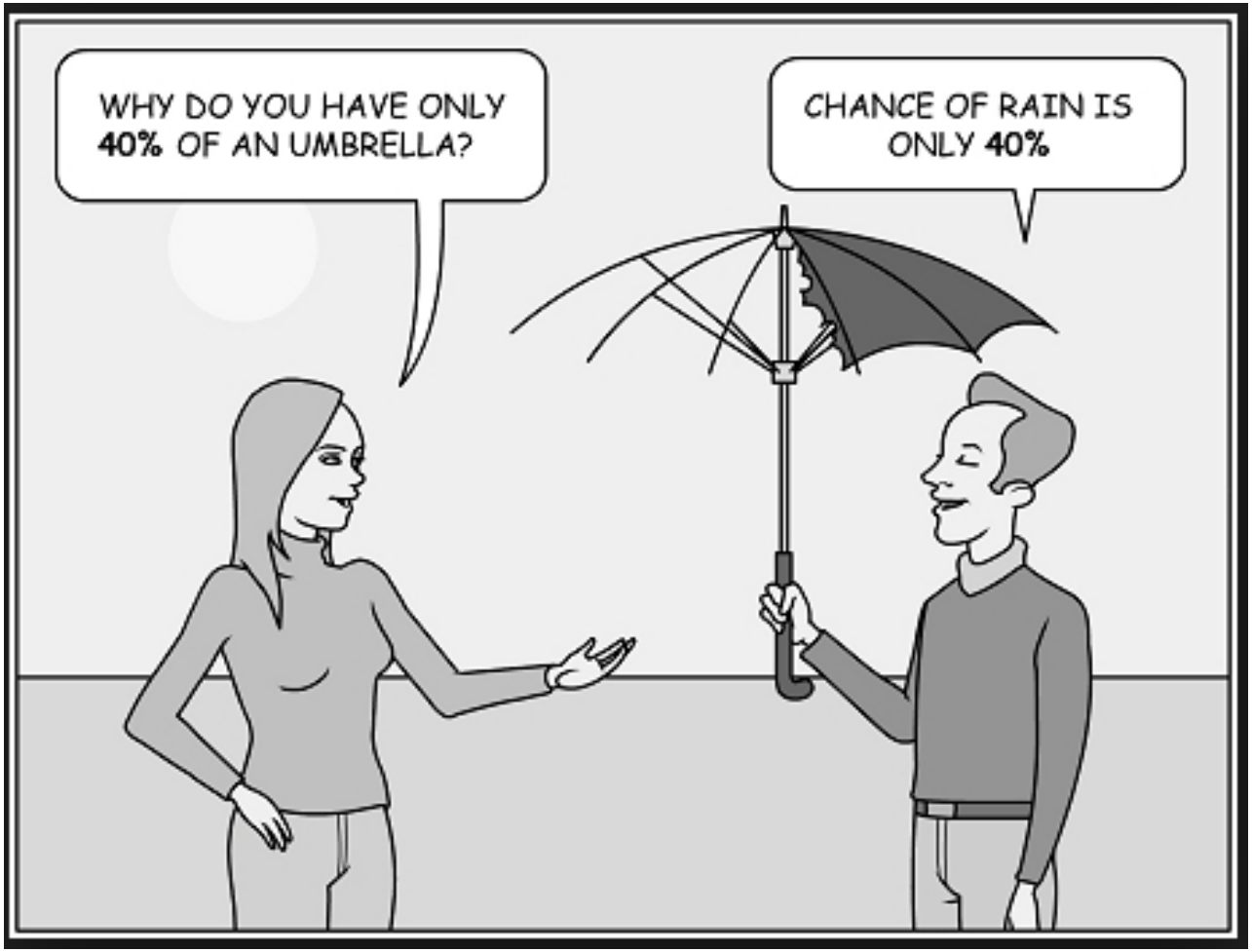
Options


WHY DO YOU HAVE ONLY 40% OF AN UMBRELLA?
CHANCE OF RAIN IS ONLY 40%

Overview

- Financial options are one of the most important and actively traded financial assets and important tools for corporate financial managers.
- We can think of the capitalization of the firm itself—that is, its mix of debt and equity—as options on the underlying assets of the firm.

Option Basics

- A *financial option* is a contract that gives its owner the right (but not the obligation) to purchase or sell an asset at a fixed price at some future date.
- There are two kinds of options:
  - a *call option* gives the owner the right to buy the asset;
  - a *put option* gives the owner the right to sell the asset.
- For every owner, there is a *option writer*.
- Disney trades for \$95.72.
- You can buy the option to buy Disney for \$90.00 on or before Jul. 1, for \$10.00.
- You can buy the option to sell Disney for \$90.00 on or before Jul. 1 for \$0.24.

Understanding Option Contracts

- When the owner buys or sells a share of stock at the agreed-upon price, she is *exercising* the option.
- The price at which the holder buys or sells the shares of stock when the option is exercised is called the *strike price*.
- There are two (more) kinds of options:
  - American options allow their holders to exercise the option on any date up to and including the *expiration date*;
  - European options allow their holders to exercise the option only on the expiration date.
- Option owners are said to be *long* and option writers are said to be *short*.

Understanding Option Contracts

- Options are not free.
- When you sell an option, you get paid for it.
- The market price of the option is also called the *option premium*.

Interpreting Stock Option Quotations

- Stock options are traded on organized exchanges.
- The oldest and largest is the Chicago Board Options Exchange (CBOE).
- By convention, all traded options expire on the Saturday following the third Friday of the month.
- Stock option contracts are always written on 100 shares of stock.
- More vocabulary: an option is said to be
  - **in-the-money** if the payoff from exercising an option immediately is positive;
  - **at-the-money** if the exercise price equals the current price of the stock;
  - **out-of-the-money** if the payoff from exercising an option immediately is negative;
- The terms deep-in-the-money are commonly used deep-out-of-the-money.

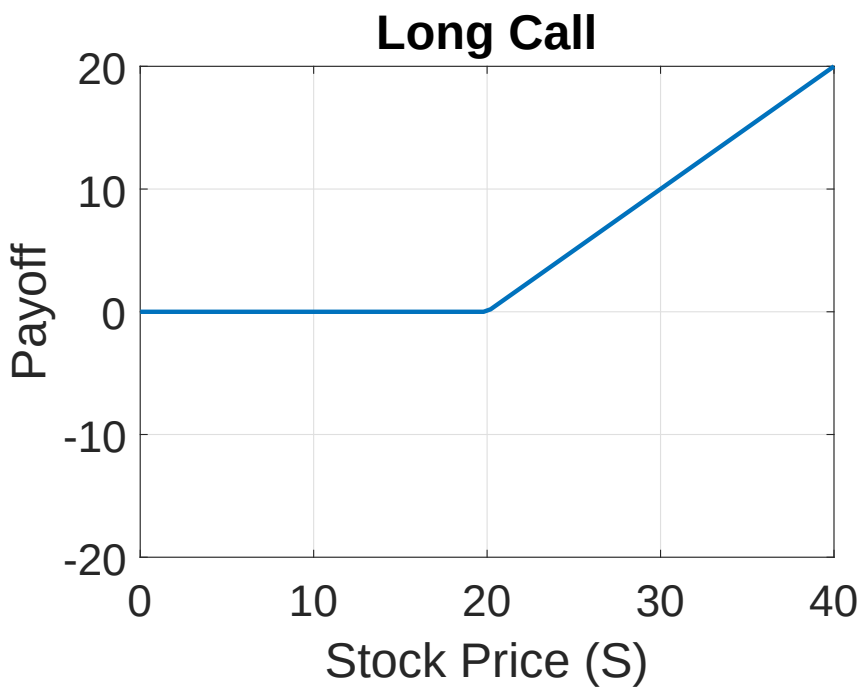
Options on Other Financial Securities

- There are options on the S&P 500, the DIJA, etc.
- Index puts allow investors to protect themselves against market downturns.
- If you use an option to reduce risk, you are said to *hedge*.
- If you use an option to bet on market movements, you are said to *speculate*.

Long Position in a Call

Let S denote the stock price at expiration, K the strike price, and C the value of the call option. The value of a **long position** in the **call option** at expiration is

$$\text{Value of Long Call} = C = \max\{S - K, 0\}.$$



Examples: Long Call

You purchase one MBI July 120 call contract (equaling 100 shares) for a premium of \$5. You hold the option until the expiration date, when MBI stock sells for \$123 per share. What is your profit/loss?

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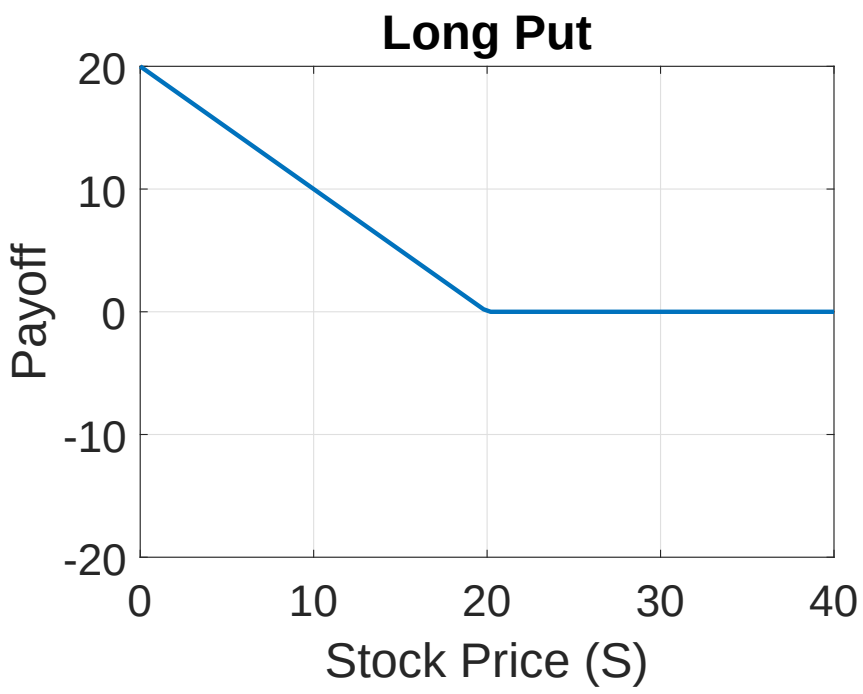
You purchase one MBI July 120 call contract (equaling 100 shares) for a premium of \$5. You hold the option until the expiration date, when MBI stock sells for \$123 per share. What is your profit/loss?

Long call profit = max{0, (\$123 - \$120)(100)} - \$500 = -\$200

Long Position in a Put

Let S denote the stock price at expiration, K the strike price, and P the value of the put option. The value of a **long position** in the **put option** at expiration is

$$\text{Value of Long Put} = P = \max\{K - S, 0\}.$$



Examples: Long Put

You purchase one MBI July 120 put contract (equaling 100 shares) for a premium of \$3. You hold the option until the expiration date, when MBI stock sells for \$123 per share. What is your profit/loss?

Examples: Long Put

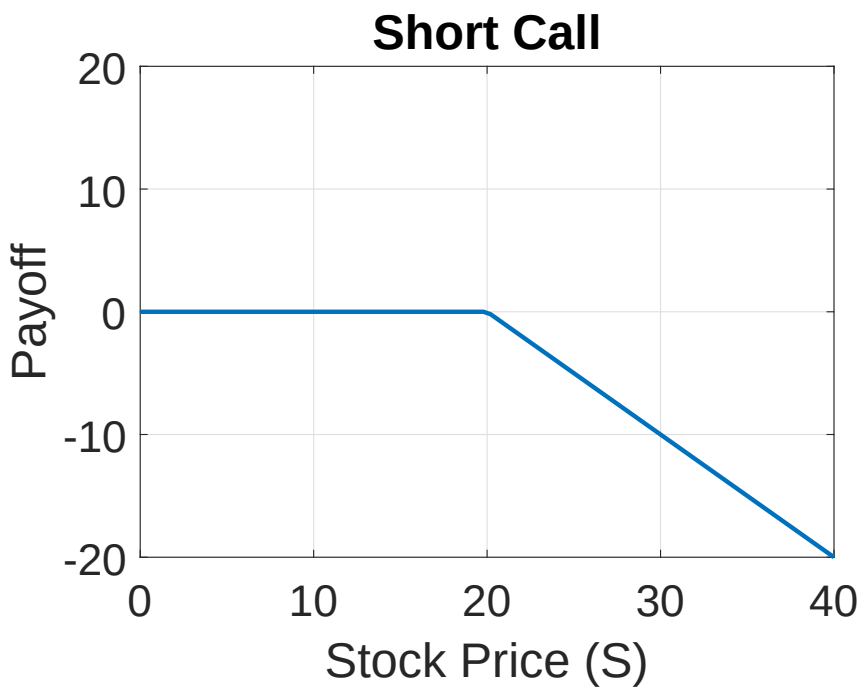
You purchase one MBI July 120 put contract (equaling 100 shares) for a premium of \$3. You hold the option until the expiration date, when MBI stock sells for \$123 per share. What is your profit/loss?

Long put profit = max{0, (\$120 - \$123)(100)} - \$300 = -\$300

Short Position in a Call

Let S denote the stock price at expiration, K the strike price, and C the value of the call option. The value of a **short position** in the **call option** at expiration is

$$\text{Value of Short Call} = -C = -\max\{S - K, 0\}.$$



Examples: Short Call

You write one MBI July 120 call contract (equaling 100 shares) for a premium of \$4. You hold the option until the expiration date, when MBI stock sells for \$121 per share. What is your profit/loss?

Examples: Short Call

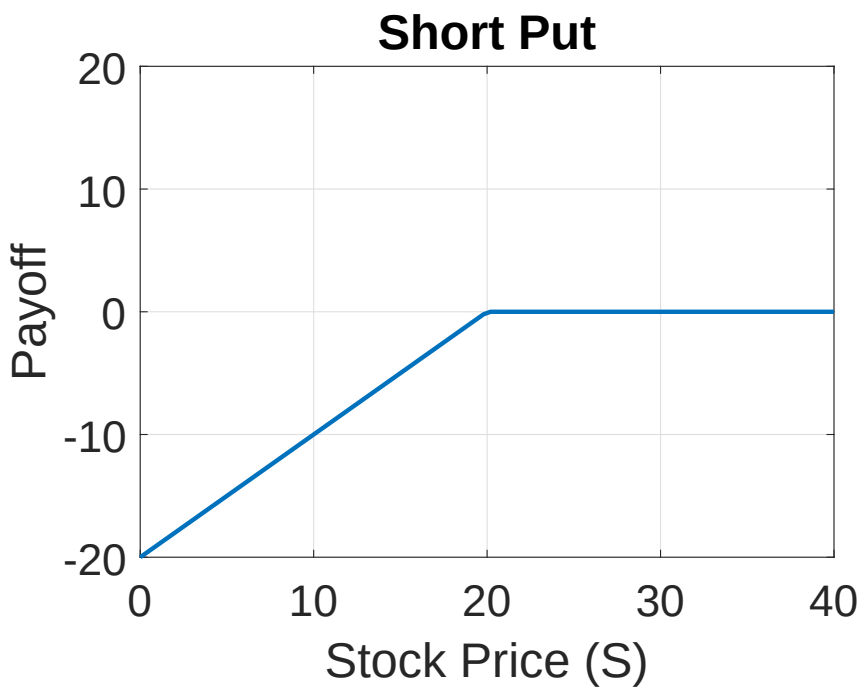
You write one MBI July 120 call contract (equaling 100 shares) for a premium of \$4. You hold the option until the expiration date, when MBI stock sells for \$121 per share. What is your profit/loss?

Short call profit = min{0, (\$120 - \$121)(100)} + \$400 = \$300

Short Position in a Put

Let S denote the stock price at expiration, K the strike price, and P the value of the put option. The value of a **short position** in the **put option** at expiration is

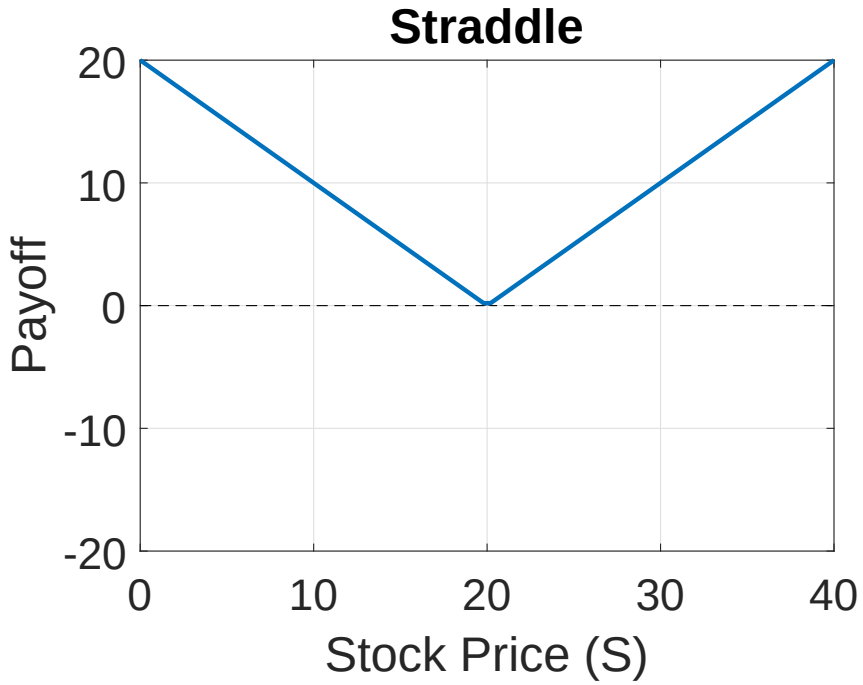
$$\text{Value of Short Put} = -P = -\max\{K - S, 0\}.$$



The Straddle

Let S denote the stock price at expiration, K the strike price, C the value of the call option, and P the value of the put option. A **straddle** is **long** a **call** and **long** a put with the same strike. Its value at expiration is

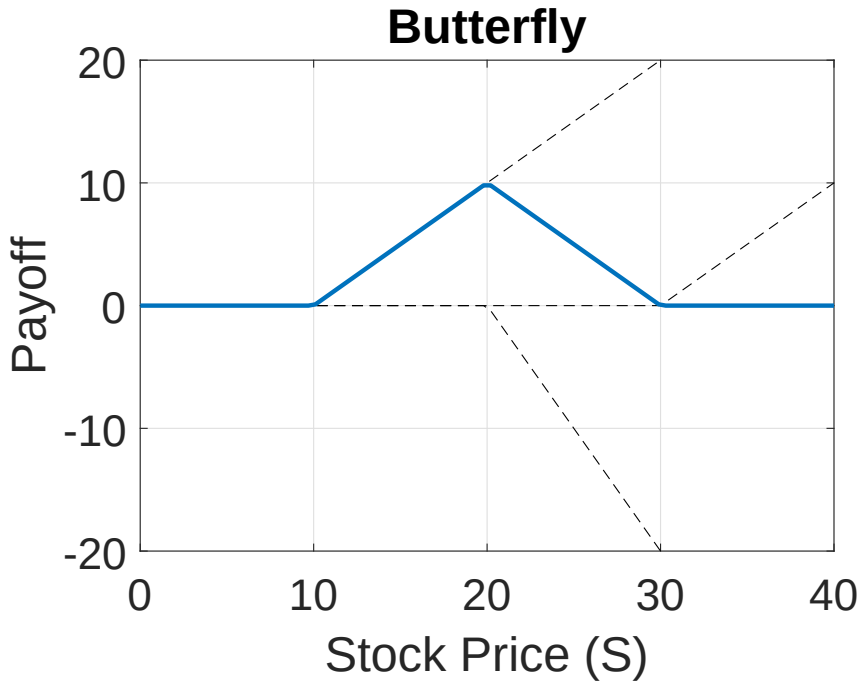
$$\text{Value of Straddle} = C(S;K) + P(S;K) = \max\{S-K, K-S\}.$$



The Butterfly

Let S denote the stock price at expiration, $K_L < K_M < K_H$ strike prices, and C the value of the call option. A **butterfly** is **long** a **call** with a **low strike**, **long** a **call** with a **high strike**, and **short** two **calls** with **medium strikes**. Its value at expiration is

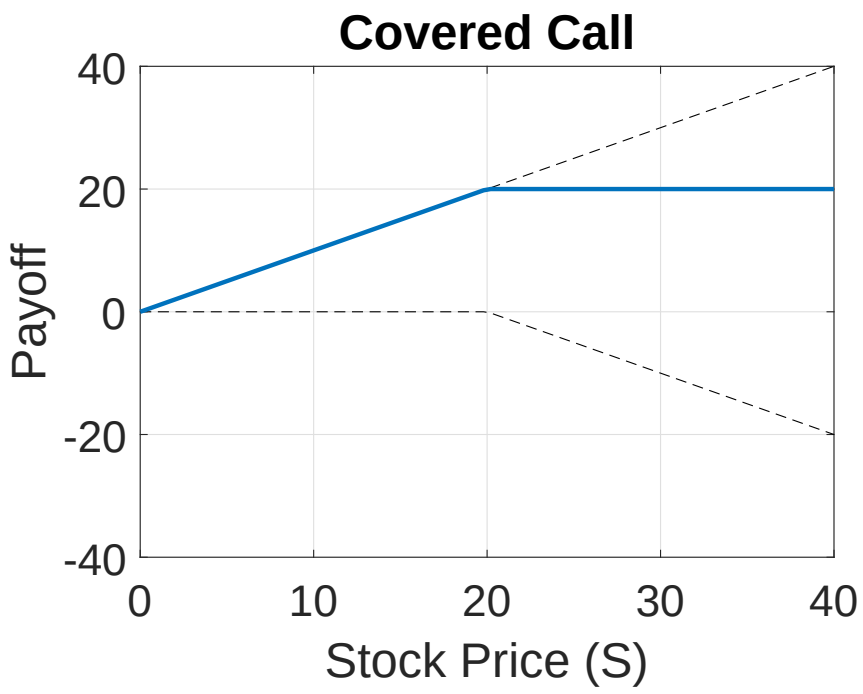
$$\text{Value of Butterfly} = C(S; K_H) + C(S; K_L) - 2C(S; K_M).$$



The Covered Call

Let S denote the stock price at expiration, K the strike price, and C the value of the call option. A **covered call** is **long** the **underlying** and **short** a **call**. Its value at expiration is

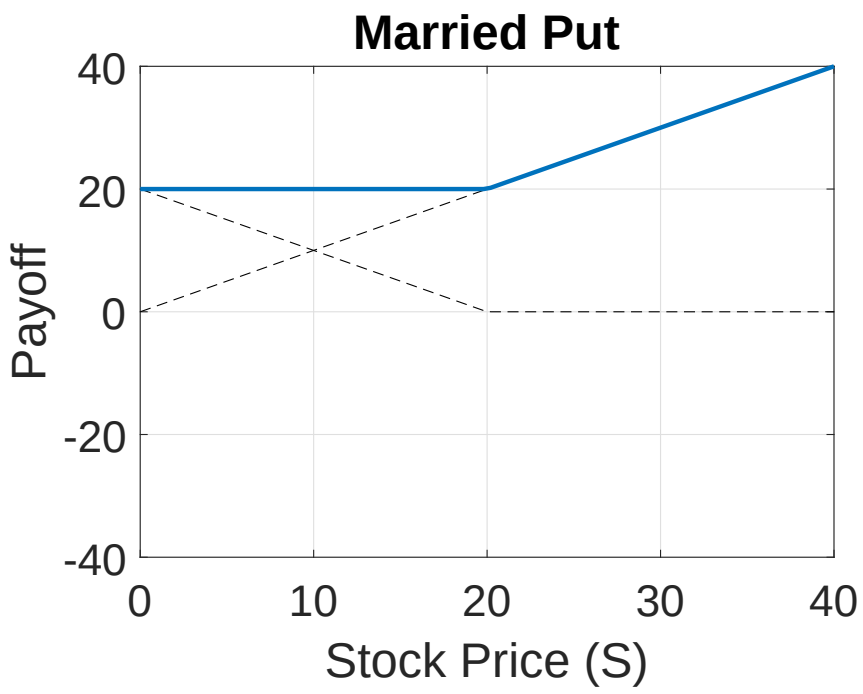
$$\text{Value of Covered Call} = S - C(S; K) = \min\{S, K\}$$



The Married Put

Let S denote the stock price at expiration, K the strike price, and P the value of the put option. A **married put** is **long** the **underlying** and **long** a **put**. Its value at expiration is

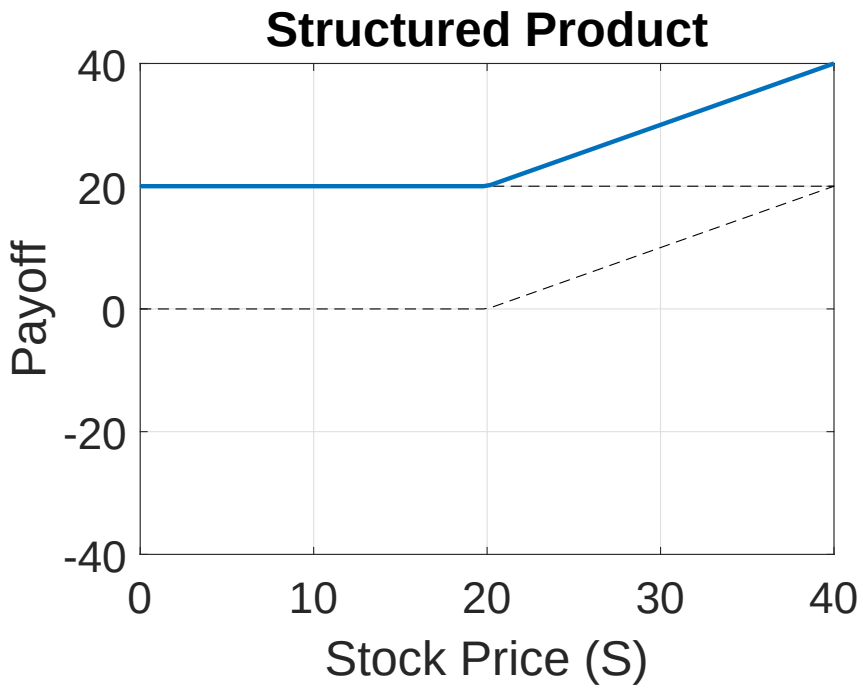
$$\text{Value of Married Put} = S + P(S; K) = \max\{K, S\}.$$



The Structured Product

The Minions' structured product was **long** the **risk-free asset** and **long** a **call**. Its value at expiration was

$$\text{Value of Structured Product} = K + C(S; K) = \max\{S, K\}.$$



Put-Call Parity

- Hmmm...the Structured Product has the same payoff as the Married Put.
- Two securities with the same cash flows must have the same price.
- **Put-Call Parity** says that

$$\overbrace{S + P}^{\text{Married Put}} = \overbrace{PV(K) + C}^{\text{Structured Product}}$$

- Put-call parity gives the price of a call in terms of the price of a put, the underlying stock, and a zero-coupon bond.

Factors Affecting Option Prices

- **Strike Price and Stock Price:** the value of a call is decreasing in the strike price (the higher the strike, the lower the value); the value of a call is increasing in the price of the stock (the higher the price of the stock, the higher the value).
- **Arbitrage Bounds on Option Prices:**
  - an American option carries all of the same rights and privileges as an otherwise equivalent European option. The American option can't be worth less.
  - a put option can't be worth more than its strike price and a call can't be worth more than the stock itself.

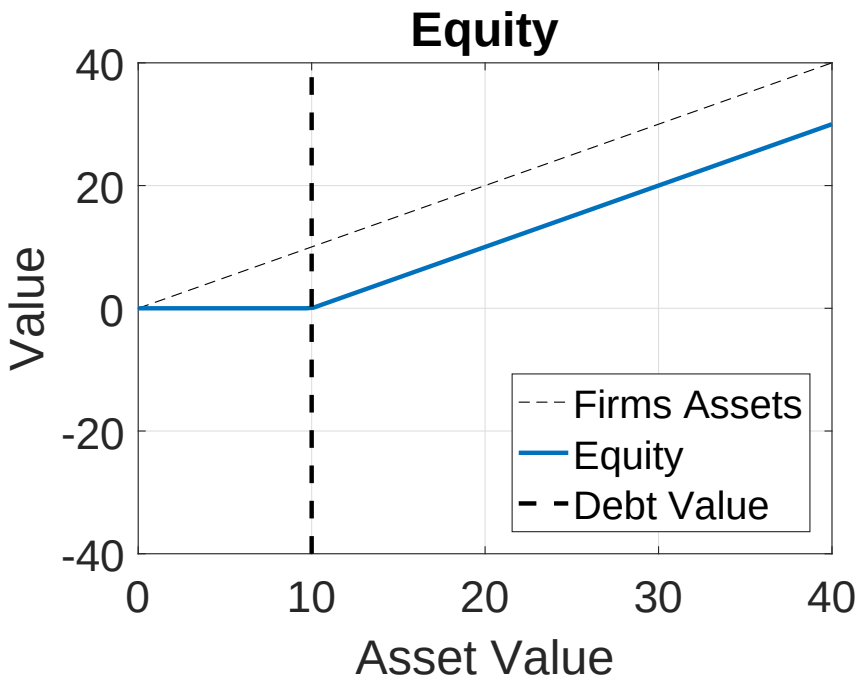
Factors Affecting Option Prices

- **Option Prices and the Exercise Date:**
  - (for American options,) the longer the time to the exercise date, the more valuable the option.
  - An American option with a later exercise date cannot be worth less than an otherwise identical American option with an earlier exercise date.
  - These arguments <u>do not</u> carry through for European options.
- **Option Prices and Volatility:**
  - The value of an option generally increases with the volatility of the stock.
  - Increasing the volatility of the stock increases the likelihood that the stock will take on extreme values (which is where options are valuable).

Equity as a Call Option

Consider a firm with \$10m in debt. It will only operate for one year. At the end of the year, its assets maybe worth anywhere from \$0m to \$40m. What's the payoff to **equityholders**?

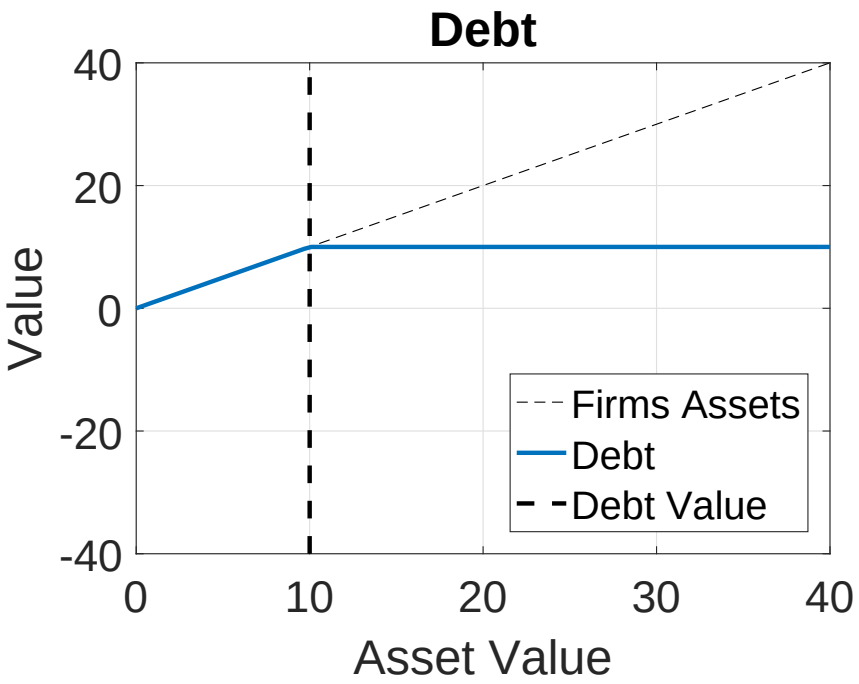
$$E = \max\{A - K, 0\} \stackrel{!}{=} \max\{A - \$10m, 0\}.$$



Debt as a Covered Call

Consider a firm with \$10m in debt. It will only operate for one year. At the end of the year, its assets maybe worth anywhere from \$0m to \$40m. What's the payoff to **debtholders**?

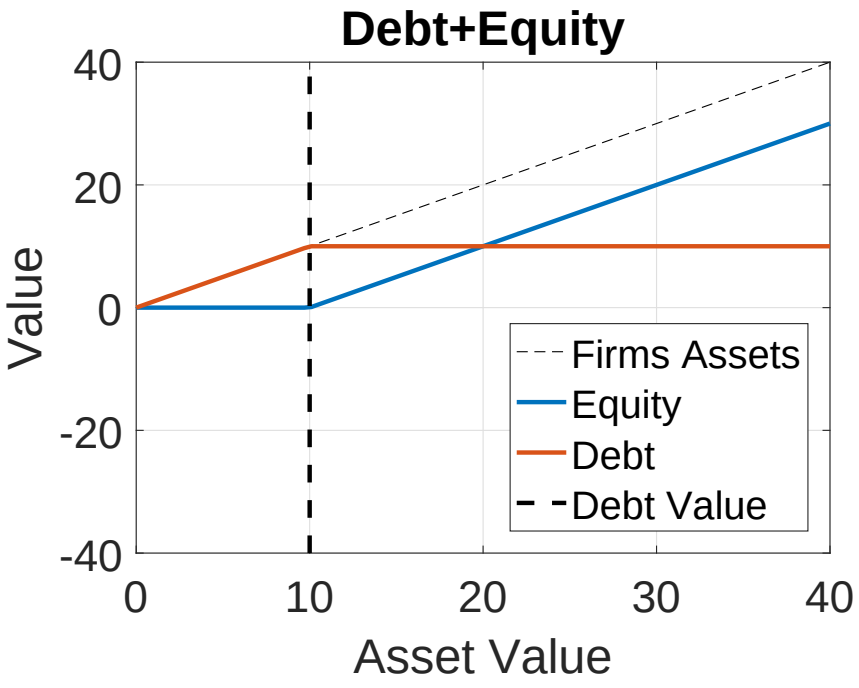
$$D = \min\{A, K\} \stackrel{!}{=} \min\{A, \$10m\}.$$



The Total Payoff

Now what's the **total payoff**?

$$E + D = \max\{A - K, 0\} + \min\{A, K\} = A$$



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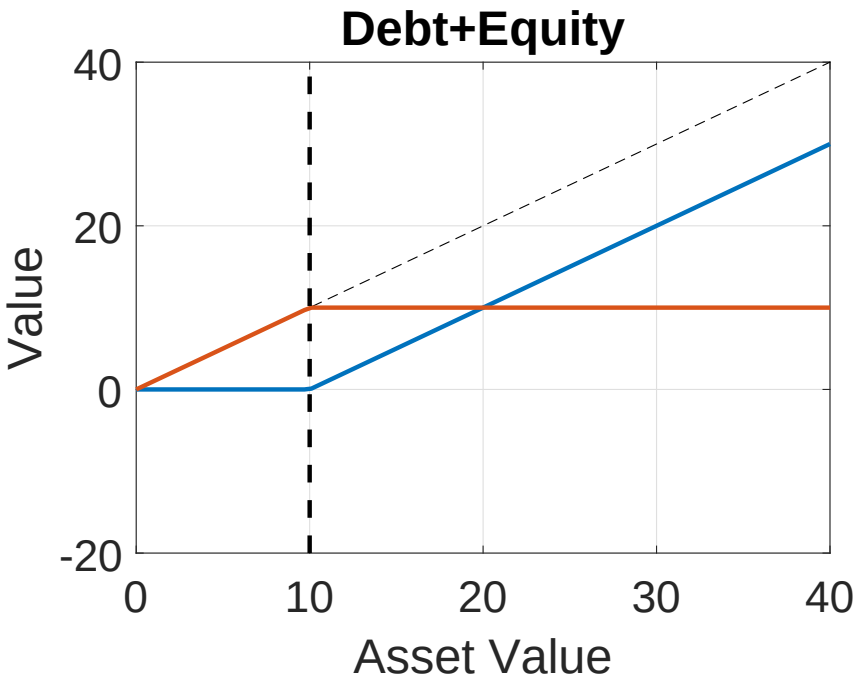

Modigliani Miller

- **Option-Speak:** a call plus a covered call equals the underlying stock

$$C + (S - C) = S$$

  regardless of the strike price.

- **Capital-Structure-Speak:** equity plus debt equals the underlying cash flows

$$E_L + D_L = V$$

  regardless of the amount of debt.

- Options provide us with an elegant proof of the Modigliani Miller Theorem, which says that the exact mix of debt and equity does not affect firm value.